

**A Flower Arrangement Event Booking and Inventory Management
System with Gemini AI-Assisted Image Analysis**

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Prepared by:

Labonite, John Lloyd,

Nadonza, Lismer John,

Pantua, Ryan,

Prudenciano, Christian Art,

Samonte, Daniela Rose

Eduardo S. Rodrigo

Capstone Adviser

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CHAPTER 1

THE PROBLEM AND ITS BACKGROUND

1.1 Project Context

The floral industry, particularly within the events sector, operates in a fast-paced and highly demanding environment where precision and timeliness are essential. Staff are required to produce flower arrangements for occasions such as weddings, corporate functions, and celebrations, all within strict deadlines. The perishable nature of fresh flowers compounds this pressure significantly, as cut flowers lose market value rapidly after harvest, necessitating constant stock replenishment to minimize waste (Maheshwari et al., 2021).

Client preferences further strain operations, as requests for rare imported varieties or locally available flowers require advance planning that manual systems cannot reliably support. Without adequate planning tools, even minor oversights can result in wilted arrangements or budget overruns. Companies that rely on fragmented, paper-based workflows such as physical notebooks, wall calendars, and informal social media messaging risk falling behind more organized competitors. These manual approaches create critical vulnerabilities including scheduling conflicts, inaccurate material estimation, undetected stock shortages, and the inability to systematically interpret client-provided inspiration images, all of which result in recurring operational delays and financial leakage.

To address these challenges, this study proposes the development of a web-based Flower Arrangement Event Booking and Inventory Management System with Gemini AI-Assisted Image Analysis. The system consolidates fragmented manual workflows into a unified digital platform that automates booking coordination, monitors real-time inventory levels, and leverages the Gemini AI API to interpret client-uploaded inspiration images and generate structured floral material suggestions. By transitioning from a manual, decentralized model to a centralized digital ecosystem, the proposed system directly addresses the operational inefficiencies identified and supports the long-term sustainability of floral enterprises within the growing events industry.

1.2 Background of the Study

Raflora Enterprises has maintained a consistent presence in the local floral industry since 2013, a strength that grants the company a stable market reputation for weddings, debuts, and corporate functions. While the company typically operates with a core staff of 10 to 15, the ability to scale to 60 personnel during peak seasons is an operational requirement. However, this capacity is currently hampered by a manual operational framework. Relying on physical notebooks, wall calendars, and fragmented social media messaging creates critical weaknesses that negate the firm's competitive advantage.

The operational lifecycle consists of two high-friction phases: Consultation and Planning, and Confirmation and Execution. In the initial phase, client preferences are manually transcribed from social media, followed by design planning where the

manual analysis of "inspiration images" results in documented delays of one to two hours per booking. This stage involves the application of an unstandardized cost-estimation using the formula: $\text{Final Price} = [(\text{Flower Cost} \times 3.5) + (\text{Hard Goods Cost} \times 2.5)] \times \text{Labor Rate}$, often involving repetitive feedback loops via chat. Once the owner and client reach an agreement on the event price and design, the owner presents a contract for the client to sign, formally confirming the booking. Raflora classifies confirmed events into three categories Advanced, Standard, and Rush each corresponding to a different scale of operation and staffing requirement.

Once confirmed, the execution phase begins with staff assignment based on the event category. Advanced events, being the largest in scale, require up to 60 personnel divided across flower arrangement, backdrop construction, wiring and lighting design, and equipment assistance. Standard events follow the same operational structure but are staffed by 10 to 20 personnel. Rush events, by contrast, are handled by 5 to 10 available staff focused primarily on flower procurement and setup, with the florist's involvement left to the owner's discretion depending on urgency.

Procurement based on visual estimation rather than inventory data; if items are unavailable, last-minute sourcing from expensive retailers is required to avoid setup delays. Finally, the logistics team executes the setup, followed by a manual Return Audit where staff count non-perishable items like vases and stand against storage records, a process that lacks formal documentation and remains prone to error.

These compounding inefficiencies ranging from the lack of a standardized pricing model to the inability to accurately track assets result in recurring record loss and financial leakage. The absence of accurate inventory records is a critical vulnerability

that has led to documented operational failures and unplanned overtime costs (Bermudez et al., 2024). Consequently, the company is losing significant productive hours that could be redirected toward high-value opportunities, such as business expansion or service innovation.

Among the available multimodal AI APIs considered for this study, the Gemini AI API developed by Google was selected for its demonstrated capability in processing both text and image inputs within a single request, its well-documented REST API structure compatible with the Laravel framework, and its accessible API tier suitable for SME-scale deployment. Unlike specialized computer vision models that require custom-trained datasets, Gemini's generative multimodal architecture allows it to interpret diverse floral arrangement styles from client-uploaded photos without prior domain-specific training, making it the most practical choice for Raflora's operational context.

This study proposes the development of a web-based Flower Arrangement Event Booking and Inventory Management System with Gemini AI-Assisted Image Analysis. While common solutions like generic inventory applications were considered, none address the specific intersection of perishable-focused planning and AI-assisted interpretation of client inspiration photos. By capitalizing on the opportunity for digital transformation, this system consolidates fragmented manual workflows into a unified digital architecture, directly addressing the operational weaknesses identified and mitigating the threats to Raflora's long-term sustainability.

1.3 Statement of the Problem

Raflora Enterprises currently manages event bookings and inventory through a fragmented manual process involving physical notebooks and social media messaging. This decentralized approach leads to documented delays of one to two hours per booking and additional setup delays of up to two hours per event due to last-minute stock shortages. To address these inefficiencies, this study develops a Web-based Flower Arrangement Event Booking and Inventory Management System with Gemini AI-Assisted Image Analysis.

Specifically, it seeks to answer the following questions:

1. What are the challenges encountered by the staff in manually coordinating client reservations, scheduling, and price reconfirmations for long-term bookings?
2. How does the lack of a structured return-tracking system for non-perishable items affect the company's asset management and inventory accuracy?
3. What difficulties do the staff face in accurately estimating floral material requirements when relying solely on client-provided inspiration images and personal experience?
4. What are the limitations of the current physical shelf-count method in detecting stock shortages prior to event preparation?
5. How does the Gemini AI-Assisted Image Analysis perform in identifying materials, and how do users rate the system's overall quality?

1.4 Objective of the Study

General Objective:

The primary objective of this study is to develop a web-based Flower Arrangement Event Booking and Inventory Management System with Gemini AI-Assisted Image Analysis for Raflora Enterprises to streamline booking coordination, automate inventory monitoring, and provide AI-driven material suggestions to minimize operational waste and processing delays.

Specific Objective:

Specifically, the study aims to:

1. To design and develop an Event Booking module that automates client reservations, scheduling, and the reconfirmation of price validity for long-term bookings.
2. To develop a Post-Event Return Tracking and Reports module to monitor the return of non-perishable assets and generate operational audit logs.
3. To integrate Gemini AI-Assisted Image Analysis that interprets client-uploaded inspiration images to generate structured floral material suggestions.
4. To implement an Inventory Management system with a real-time stock shortage detection and flagging mechanism.
5. To evaluate the functional suitability of the Gemini AI module and the system's overall usability using a standardized evaluation instrument.

1.5 Scope and Limitation

The study focuses on the design and development of a web-based Flower Arrangement Event Booking and Inventory Management System with Gemini AI-Assisted Image Analysis, specifically tailored for the operations of Raflora Enterprises. The system is designed to transition the business from a manual, paper-based workflow to a centralized digital ecosystem.

The system is accessible via standard web browsers and implements role-based access control (RBAC) for three distinct user roles:

Admin: Responsible for managing the overall booking calendar, validating AI-generated material suggestions, overseeing inventory levels, and generating operational reports.

Client: Accesses a self-service portal to submit event booking requests, upload inspiration images for AI analysis, and receive price quotations.

Staff: Utilizes the system to view assigned event checklists, update physical stock levels, and log the return of non-perishable materials.

The functional core of the system is composed of the following modules:

Event Booking Coordination: Manages the lifecycle of a reservation, from initial request and scheduling to final confirmation. It supports both pre-set packages (with automated material lists) and customized arrangements.

Gemini AI-Assisted Image Analysis: Utilizes the Gemini AI API to analyze client-uploaded inspiration photos. The AI interprets visual data to suggest a

structured list of floral materials, which serves as a professional reference for the administrator.

Inventory Tracking: Monitors real-time stock levels and flags shortages relative to confirmed bookings. The system implements a tiered alert mechanism, performing weekly inventory checks starting four weeks prior to an event and escalating to daily alerts during the final week. To address stock unavailability, the system includes a Seasonal Substitution feature that recommends alternative floral materials when requested items are out of stock or out of season, minimizing operational disruptions and unplanned procurement costs.

Post-Event Return Tracking: Tracks the recovery of non-perishable assets (vases, stands, lighting), flags missing or damaged items, and updates inventory records accordingly.

Price Validity and Reconfirmation: Manages the volatility of floral pricing by marking initial quotes as "tentative" and triggering automatic reconfirmation notifications for long-term bookings.

Reports and Audit Logs: Generates timestamped logs and dashboard analytics regarding material usage, booking history, and business performance.

Limitation

The system is delimited to the operations of Raflora Enterprises' single-branch facility and does not support multi-location synchronization. The Gemini AI-assisted image analysis is implemented as a decision-support tool; it provides suggestions

based on visual patterns, but the final validation and authorization of material lists remain the sole responsibility of the Administrator. Consequently, the system is not responsible for the absolute botanical accuracy of the AI's suggestions, as these are constrained by the quality, lighting, and clarity of the client-uploaded images.

Technical limitations include a mandatory requirement for continuous internet connectivity, as the system relies on cloud-based hosting and the Gemini AI API. The system does not support offline mode or a dedicated native mobile application. Finally, the project evaluation is restricted to usability and functional testing among the internal staff and administrators of Raflora Enterprises. Furthermore, the collection and storage of client-uploaded images and personal booking information are handled in accordance with the principles of Republic Act No. 10173, also known as the Data Privacy Act of 2012, ensuring that all user data is accessed only by authorized system roles and is not shared with third parties.

1.6 Significance of the Study

The development and implementation of this system are expected to provide significant operational and technical value to the following stakeholders:

Administrators /Management: The system provides a professional decision-support tool that eliminates the critical business risk of relying on personal memory or unofficial communication channels like social media messaging. By centralizing booking schedules, integrating automated inventory alerts, and shifting from a fragmented manual workflow to a centralized digital ecosystem, the organization achieves a higher level of operational maturity. This digital transformation allows management to shift their focus from manual data entry to strategic quality control, enabling the business to scale operations during peak seasons without a proportional increase in administrative burden.

Clients: Clients benefit from a streamlined self-service portal that facilitates faster price quotations and more accurate interpretations of their inspiration images. The Price Validity and Reconfirmation mechanism ensures transparency regarding the volatility of floral pricing, fostering trust through automated notifications for long-term bookings.

Future Researchers: This study serves as a practical reference for implementing generative multimodal architecture within SME-scale deployment. It provides a framework for how the Gemini AI API can be utilized as a decision-support tool to solve domain-specific challenges in industries dealing with perishable-focused planning and inventory monitoring.

Staff: The system provides immediate operational relief by replacing the current high-friction phases of manual coordination with a streamlined digital workflow. By utilizing Gemini AI-Assisted Image Analysis, staff are no longer burdened by the documented delays of one to two hours typically spent on the manual transcription and visual analysis of client photos. Additionally, the transition from the unreliable physical shelf-count method to a structured Post-Event Return Tracking module reduces the risk of manual errors and prevents the unplanned overtime costs often caused by asset recovery disputes.

1.7 Definition of Terms

The following terms are defined operationally as they are used and interpreted within the context of this study and the operations of Raflora Enterprises:

Event Booking: The process of receiving, scheduling, and confirming client event reservations through the system, matched against available inventory and the event calendar.

Flower Arrangement: refers to the structured composition of floral and decorative materials prepared for a specific client event, serving as the primary service output that the system's booking, inventory, and AI analysis features are designed to support.

Gemini AI-Assisted Image Analysis: The integration of the Gemini AI API into the system to process client-uploaded inspiration images and translate visual data into a structured list of suggested floral materials.

Inspiration Image: refers to a client-provided photo showing desired event themes, color schemes, or arrangement styles, uploaded through the system for AI-assisted analysis.

Inventory Management: refers to the process of recording, tracking, and monitoring floral materials and supplies specific to Raflora Enterprises confirmed and upcoming event bookings, including stock level monitoring against booking requirements and the recording of material release.

Post-Event Return Tracking: The mechanism used to monitor the return of non-perishable assets (such as vases, stands, and lighting equipment) following an event and flagging missing or damaged property for administrative follow-up.

Pre-set Package: refers to a predefined floral arrangement bundle with fixed material lists and pricing that clients may select during the booking process as an alternative to a fully customized arrangement request.

Price Validity and Reconfirmation: The system process that marks initial quotations as "tentative" and generates notifications to re-verify pricing and material availability prior to the event date.

Seasonal Substitution: refers to the recommendation of alternative floral materials when requested items are unavailable or out of season.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Introduction

This chapter presents a comprehensive review of related literature and studies essential to the architectural and functional design of the Flower Arrangement Event Booking and Inventory Management System with Gemini AI-Assisted Image Analysis. Unlike traditional retail systems that manage non-perishable hardware, the floral industry faces a unique dual challenge: the high perishability of inventory and the extreme visual complexity of client requirements. This review establishes the technical necessity for a system that can bridge the gap between unstructured visual inspiration and precise material quantification.

The literature surveyed from both local and foreign sources explores the evolution of perishable inventory management, the logistics of event booking coordination, and the emerging role of AI-assisted tools for Small and Medium Enterprises (SMEs). Following this, a review of related studies evaluates existing web-based platforms for event and reservation management, highlighting their strengths and their failure to address the translation gap between reference images and stock deduction. By synthesizing these findings, this chapter establishes the Research Gap that the proposed system seeks to fill and presents the Conceptual Framework using the Input-Process-Output (IPO) model to illustrate the system's operational logic within the specific context of Raflora Enterprises.

2.2 Review of Related Literature

Local Literature

Oyangoren and Camilo (2025) examined AI-tool adoption among administrative personnel in Philippine organizations, using interviews and thematic analysis to show how AI automates routine tasks but faces barriers such as inadequate infrastructure and resistance to change. They emphasize that training and organizational support are critical for successful adoption, which is relevant to Raflora's Gemini AI-Assisted Image Analysis module and its implementation among non-technical staff. Their findings support the need for user oriented design and in house training in the proposed system.

Balba, Junio, and Sanares (2024) analyzed how AI technologies improve customer experience and business performance in Philippine SMEs, combining qualitative and quantitative methods to assess machine-learning models, predictive analytics, and automated service systems. Their results show that digital AI tools enhance efficiency and customer satisfaction, though data privacy and technical skill gaps remain, which supports integrating AI-assisted features into Raflora's web-based platform. This work justifies AI-assisted decision support for a service oriented SME seeking digital transformation.

Sarucam, Paglinawan, Paglinawan, and Otto (2023) developed a QR-integrated web application for small scale businesses, using an Agile-based approach to allow customers to check store and product availability online. Usability surveys showed

that users find the system useful and easy to use, supporting the design of a responsive web interface for small Philippine enterprises similar to Raflora's needs. Their work informs the proposed web-based, QR-ready interface for client and staff access.

Jiang and Denuuevo (2024) studied digital transformation strategies of commercial banks and SME financing, using empirical bank and loan data to show how big data risk control and blockchain based finance improve SME access to credit. They highlight scenario based service design and digital readiness gaps, which reflect the broader context Raflora faces when adopting a web-based management system as an SME. Their findings support the need for simple, scenario driven system features and organizational preparedness.

Ortega-Dela Cruz and Dela Cruz (2025) examined AI integration in IT education in the Philippines through a survey of IT students and educators, applying the Technology Acceptance Model. They identified key challenges (inadequate infrastructure, limited AI literacy, ethics/privacy concerns, rural connectivity gaps) alongside opportunities such as personalized learning, automated administrative tasks, and enhanced data analysis. The study recommends curriculum updates, policy frameworks, and industry collaboration to overcome barriers. These insights are highly relevant to Raflora's overall system, especially the Gemini AI-Assisted Image Analysis and inventory modules, as they stress the need for infrastructure support, ethical guidelines, and educator/staff training to successfully deploy AI-enhanced booking and management tools in Philippine settings.

Albano et al. (2025) examined Generation Z Filipinos' perceptions of text-generative AI using the Unified Theory of Acceptance and Use of Technology (UTAUT 2) extended with ethical considerations and multiple regression analysis on student respondents from the University of Santo Tomas. They identified Effort Expectancy, Performance Expectancy, Facilitating Conditions, Hedonic Motivation, Habit, Ethical Considerations, Perceived Risk, Attitude, Region, and Gender as significant predictors of behavioral intention to adopt AI. The study highlights infrastructure, cost, and ethical barriers in the Philippine education context while noting widespread student use (83% for research/writing). Their findings reinforce the importance of addressing facilitating conditions and ethical factors when implementing Raflora's Gemini AI-Assisted Image Analysis, particularly for younger users or staff involved in event booking and inventory decisions.

Foreign Literature

Maheshwari, Singh, and Tiwari (2021) developed an IoT-based cost-benefit model for perishable inventory in MSME food retail, using fractional programming and particle swarm optimization to show how automated monitoring reduces spoilage and inventory costs. Their findings indicate that real-time stock tracking and optimized ordering are feasible for small scale enterprises, which supports the need for an automated Inventory Management System in Raflora to reduce manual counting errors and last-minute high cost purchases. This work is relevant to the system's event linked shortage detection and inventory alert mechanisms for perishable floral materials.

Momotaz and Banik (2021) analyzed seasonal price and wastage patterns in fresh flower businesses in Bangladesh, using quantitative price monitoring and season based demand supply analysis to show how shortages and oversupply affect profitability and post-harvest losses. They found that rainy season shortages drive price spikes while spring over supply depresses prices, which underpins Raflora's planned Seasonal Substitution and Emergency Substitute module for peak event months. Their work directly supports the integration of seasonal aware rules into an SME floral event context.

Macías-López, Mendoza, and González (2021) formulated a multivariate inventory model for perishable items that links price, stock level, and freshness state, using optimization routines to derive order quantities, cycle times, and pricing strategies under varying shelf-life conditions. Their results show that extending shelf life improves profitability and that dynamic pricing outperforms static rules, which supports the rationale for Raflora's Price Validity and Reconfirmation and AI-Suggested Materials modules. These concepts help justify price and freshness linked ordering logic in the proposed system.

Kusumawardhani et al. (2025) developed a web-based reservation system for catering MSMEs using Agile-Scrum, incorporating real-time availability and automated coordination to reduce manual booking delays and scheduling conflicts. User acceptance testing showed that the system improves customer satisfaction and service efficiency in small service operations, which directly informs Raflora's Event Booking Coordination module. Their work supports the design of real-time availability and role-based booking features in the proposed system.

Rao (2026) explored AI-driven real-time inventory management for perishable goods, using machine learning forecasting, computer vision quality assessment, and IoT sensors to reduce spoilage and optimize storage. The study demonstrates how predictive analytics and freshness monitoring improve inventory accuracy and operational performance in perishable goods supply chains, which provides a theoretical foundation for Raflora's Gemini AI-Assisted Image Analysis and Perishable Inventory Management modules. This literature supports the integration of real-time data and image-based quality checks for floral materials.

Ilcheva (2025) conducted a comprehensive literature review of six AI-based models addressing sustainable inventory management in grocery retail, focusing on demand forecasting, inventory control, and perishables classification. Her Aalto University thesis demonstrates how machine learning reduces food waste by 20-49% through predictive analytics, directly paralleling Raflora Enterprises' challenges with rapidly perishing cut flowers during peak wedding seasons. These findings validate the integration of Gemini AI-driven forecasting and real-time stock optimization in Raflora's proposed system to eliminate shortages and over-procurement losses.

Kumar (2025) reviewed and synthesized advanced optimization models (FIFO/LIFO, Just-In-Time, multi-echelon inventory, dynamic pricing, and AI/ML demand forecasting) tailored for time-sensitive retail perishables, explicitly citing the floral industry as a prime example where short shelf life and freshness requirements drive high waste. The study showed that combining IoT cold-chain monitoring, blockchain traceability, and machine-learning-based replenishment can cut expired stock by up to 30% and reduce transport costs while maintaining service levels. This

work strongly supports your Raflora's system's inventory management and event-booking features: the dynamic pricing and automated replenishment strategies justify the Price Validity/Reconfirmation modules, and the floral-specific examples reinforce the need for real-time stock visibility to prevent double-booking or out-of-stock issues during large event arrangements.

Sundaravadivel et al. (2025) proposed Ros-AI, a multimodal framework combining UAV/drone imaging, deep learning, unsupervised computer vision, and large language models (LLMs) for automated flower detection, counting, and inventory management in ornamental nurseries. The system achieves high accuracy in real-time stock tracking and quality assessment while integrating generative AI for decision support. Results showed significant improvements in inventory accuracy and reduced manual labor for perishable ornamental crops. This directly relates to your Raflora's Gemini AI-Assisted Image Analysis and inventory management features: the multimodal image analysis and LLM-enhanced suggestions mirror AI-Suggested Materials and event-booking stock validation, while addressing freshness and availability challenges in floral systems.

Mantilla et al. (2025) conducted a systematic review of Industry 4.0 technologies (IoT, AI, computer vision, and machine learning) in the global export flower sector, showing how AI optimizes resource management, predicts disease outbreaks, and enables real-time inventory and supply chain visibility to cut spoilage and improve sustainability. Case examples from major exporters demonstrated 15–25% reductions in waste through predictive analytics and automated quality grading via computer vision. This work strongly justifies your system's Raflora's perishable inventory

backbone, event booking reliability (preventing overbooking of short-shelf-life stock), dynamic Price Validity/Reconfirmation modules, and Gemini AI image analysis for assessing flower quality and suggesting complementary materials in real time.

Wani et al. (2023) reviewed robotics, computer vision, and machine learning applications in floriculture, including automated harvesting, disease detection, and robotic systems for flower arrangement. They highlighted QuickFlora's FloraBot prototype, which uses computer vision and ML algorithms to recognize flower types/colors and autonomously assemble arrangements, along with image-processing robots for precise handling of delicate cut flowers. The study stressed labor shortages and the need for scalable automation to maintain quality in high-volume production. These innovations directly support the Raflora's Gemini AI-Assisted Image Analysis component of your system (real-time flower recognition and arrangement suggestions during event booking) and the overall inventory management for perishable items, enabling AI-Suggested Materials and automated quality checks to reduce spoilage and support reliable event scheduling.

2.3 Review of Related Studies

Local Studies

Tanaman, Baylosis, Abiles, Catungal, and Encarnacion (2023) developed a web-based inventory management system for a small business enterprise in Pagadian City, Zamboanga del Sur, using a web application framework to automate stock recording and reporting. The system showed that online inventory monitoring reduces manual data entry errors and improves operational efficiency, which directly supports Raflora's current need to replace paper based inventory processes. Their work is a relevant local precedent for an SME focused web-based inventory module.

Espenida and Formanes (2023) developed an e-inventory management system for the supply office of secondary schools in Sorsogon, Philippines, using a web-based approach to enable remote access and inter-departmental coordination. Evaluation showed that the system Improves information flow and reduces manual follow ups, which supports the idea of an online inventory structure for Raflora that links event requests with stock availability. This study underpins the need for real-time inventory visibility in a service oriented SME.

Calucin, Hapa, and Benito (2024) developed a web-based clinic records and inventory management system for St. Paul University Quezon City, integrating patient records, inventory tracking, and appointment scheduling. The system met software quality standards and user acceptance metrics, demonstrating that a single platform can combine records and inventory operations, similar to Raflora's

integrated event booking and inventory needs. Their work supports the design of a unified system that links bookings and operational workflows.

Vega and Generoso (2023) created an online facility reservation system for Lyceum of the Philippines University–Batangas using PHP and MySQL, focusing on real-time availability checking and automated confirmations to reduce scheduling errors. User testing showed that the system improves notification efficiency and administrative productivity, which directly informed the design of Raflora’s Event Booking Coordination module. Their work supports the use of conflict-checking and automated confirmation features in the proposed system.

Reyna (2023) examined the development of FlowerHub, an online flower shop system for local flower shops in Surigao City using object-oriented analysis and the Laravel PHP Framework. Through use-case modeling, class diagrams, and System Usability Scale (SUS) evaluation (score: 85/100), the study demonstrated how a web-based platform automates manual ordering, real-time inventory tracking (add/update/remove products and quantity in stock monitoring), order processing, and sales reporting. It addresses perishable goods challenges such as limited reach and inefficient stock management. These findings directly support Raflora’s flower arrangement event booking and inventory management modules by providing a proven local model for digital order fulfillment and centralized inventory control that can be extended with Gemini AI image analysis for arrangement recommendations.

Valdez et al. (2025) analyzed the integration of AI tools among Filipino college students and young professionals, noting a significant trend toward using Large

Language Models (LLMs) like Gemini for "process automation" and "content generation." They argued that while technology spreads rapidly, successful implementation requires a clear understanding of the tool's capabilities to avoid "policy gaps." This research is applicable to your capstone Raflora's as it highlights the necessity of a well defined AI workflow ensuring that your system's image analysis correctly translates visual flower arrangements into inventory data without technical friction.

Pandramisu (2023) developed an event management system as a graduate capstone project at Governors State University, providing a platform for users to view past and upcoming events and manage event logistics through a web application. The system adopted Agile web development methodology to support rapid iteration and quality assurance. Through an international study, Raflora references its structured event lifecycle model from creation to post-event reporting which mirrors the proposed system's event booking lifecycle for floral arrangement clients.

Coro, Cuenca, and Ramos (2022) developed an inventory management system for the finished goods of RISO Packaging Technology Corp. at Batangas State University JPLPC–Malvar, addressing discrepancies between manual inventory counts and actual stocks. The system automated stock tracking, generated inventory reports, and minimized human error from manual counting. Raflora's Inventory Management Module for floral materials and arrangement supplies was informed by this study's emphasis on real-time stock monitoring and automated discrepancy detection.

Atilano-Tang, L. A., and Damsani, K. (2023) assessed the quality of the inventory management system of the BARMM-Ministry of Public Works-Basilan District Engineering Office in the Philippines, examining how system quality affects organizational effectiveness and efficiency in procurement and supply chain management. The study found that well-implemented inventory systems significantly enhance government service delivery, and recommended strategies to improve data accuracy and procurement transparency. This is relevant to the proposed system's Raflora's need to ensure accurate, real-time inventory records for flower stock and event supplies.

Ahmad (2023) developed and evaluated the e-AIMSS, an electronic asset inventory and management system deployed in a school setting in the Philippines, resulting in measurable improvements in resource optimization and organizational productivity. The study established that digital inventory systems significantly outperform manual methods in tracking school assets, with reduced discrepancies and faster retrieval. Raflora's inventory system for floral arrangement supplies and materials adopts similar asset-tracking principles to ensure accurate stock levels and reorder alerts.

Weng and colleagues (2024) applied a mixed-methods Design Thinking approach to determine user requirements for a tailored web-based inventory and sales management system for micro-small construction supply stores in the Philippines, presented at the World Symposium on Software Engineering 2024. The study employed Swimlane Diagrams and SWOT analysis and found that business owners

need real-time, intuitive inventory dashboards tailored to their specific trade. Raflora applies the Design Thinking philosophy from this study in crafting a user-centered interface for floral inventory management.

Onsay and Rabajante (2024) from the University of the Philippines Los Baños proposed using a combination of Participatory Rural Appraisal (PRA) and Machine Learning (ML) to forecast natural disasters in the Bicol region, representing one of the early published local applications of AI/ML in community decision-making. The study underscores the growing viability of AI-driven analytical tools in Philippine contexts. Raflora references this study as evidence of expanding AI adoption in local academic research, providing academic grounding for Raflora's own AI-assisted image analysis feature.

Foreign Studies

Deepthi et al. (2025) developed a web-based event booking platform using the Django framework and PostgreSQL database to streamline event management for both organizers and attendees, incorporating real-time booking, ticketing, and payment integration. Usability testing confirmed that the centralized platform reduces manual coordination effort and scheduling conflicts while supporting large scale event operations. The role-based booking tools and confirmation logic demonstrated in this study directly inform the design of Raflora's Event Booking Coordination module.

Nguyen et al. (2025) proposed the Automated Image Recognition (AIR) Framework, using generative AI to synthesize annotated datasets and train deep-learning models for image recognition. Evaluation showed that AIR improves dataset quality and classification performance, which is directly relevant to Raflora's Gemini AI-Assisted Image Analysis module for automating object recognition behavior from client images. This framework supports the idea of AI-assisted generation of recognition like rules.

Patil and Rana (2023) reviewed the use of machine learning and deep-learning techniques in computer vision, including object detection and image classification, using recent literature surveys to show how AI-powered systems achieve high object identification accuracy. Their review supports the use of similar image analysis principles in Raflora's Gemini AI module for Interpreting floral arrangement photos. The work underpins the selection of object recognition and classification techniques for clients submitted images.

Vimalarani and Sathiyaseelan (2021) developed an event-management system for conferences and workshops using JavaServer Faces and MySQL, incorporating registration, scheduling, and profile management modules. Testing showed that the centralized platform reduces manual scheduling errors and improves coordination, which aligns with the proposed event booking and scheduling structure for Raflora Enterprises. Their system demonstrates how a web-based platform can manage event related workflows in small scale settings.

Li, H., Zhang, Y., and Wang, F. (2024) proposed an intelligent flower-arrangement assistance system that performs image-based recognition of floral materials, matches them with database-stored designs, and guides manual or automated arrangement. Their results demonstrate that target-detection and data-matching algorithms can significantly speed up the design process while maintaining aesthetic quality, which supports Raflora's Gemini AI-Assisted Image Analysis and AI-Suggested Materials modules by validating the use of visual recognition for selecting appropriate flowers from available stock.

Parves, KM et al. (2024) proposed an AI-powered inventory management system using YOLO (You Only Look Once) object detection for real-time identification and tracking of warehouse items, achieving over 90% accuracy in differentiating packages and extracting serial numbers for tracking. The system automated data collection, reduced manual labor, and enabled seamless integration with existing warehouse management software. This research strongly supports Raflora's image-based inventory identification module, demonstrating that real-time AI object detection can accurately recognize and update stock levels for physical items analogous to identifying specific flower types from visual input.

Cayli, O. and Oralhan, Z. (2024) investigated the application of AI algorithms including Gradient Boosting Machines and Support Vector Machines to enhance inventory management for retail SMEs. Using historical sales data, the study trained models to answer questions about weekly sales targets under seasonal and promotional influences, achieving more accurate stock-level determination than manual methods. This study supports Raflora's inventory module by demonstrating

that AI-driven demand forecasting can help florist businesses avoid stockouts and reduce waste for time-sensitive floral materials

JosephNg, P.S., Al-Sofi, S.M., Phan, K.Y., Lim, J.T., and Lai, S.C. (2023) designed a web-based platform for event-venue booking and management, demonstrating how digital platforms can handle multiple tasks such as financial management, scheduling, and coordination for events. Their study showed that a properly structured web system improves event organization efficiency and reduces human error in managing reservations. This is directly relevant to the Flower Arrangement Event Booking System module of the capstone, as it validates the use of a web-based platform for managing event bookings, client scheduling, and resource tracking in a centralized digital environment.

Prasasti, K.F.A., Anansa, M.A., Putri, L.M.N., and Lestari, D.I. (2025) designed a web-based marketplace platform called FlowerNest for micro, small, and medium enterprise (MSME) florists in Indonesia, using the Eisenhower Matrix, MoSCoW Framework, and User Journey Map to define system features. Their usability testing confirmed functional feasibility and positive user acceptance, with plans to expand to AI-based recommendations and AR previews. This study supports the capstone's approach of building a flower-specific web platform with AI-assisted features, validating that florist-oriented digital systems with intelligent recommendation capabilities are viable and well received by users.

Onyejiaku, T., Nwakanma, C., and Ekoema, B. (2024) developed a deep learning-based web application capable of recognizing flower species and monitoring

their status using convolutional neural network models. Their results showed that deep learning significantly improves the accuracy of real-time flower identification in a web-deployed environment, enabling automated monitoring of floral health. This study directly supports the capstone's Gemini AI-Assisted Image Analysis module by proving that AI-powered image recognition can be successfully integrated into web platforms to identify and analyze floral materials automatically.

2.4 Synthesis of the Review of Related Literature and Studies

The reviewed literature collectively establishes a strong technical and operational foundation for the proposed system by converging on three major themes: the necessity of automated perishable inventory management, the effectiveness of web-based event booking platforms, and the growing viability of AI-assisted image analysis in specialized service industries.

On the subject of perishable inventory, studies by Maheshwari et al. (2021), Macías-López et al. (2021), Ilcheva (2025), and Kumar (2025) consistently demonstrate that manual stock management is fundamentally inadequate for time-sensitive perishables such as cut flowers. These works collectively support the need for real-time, event-linked inventory tracking with automated shortage alerts precisely the function addressed by the system's Inventory Management module. The floral-specific findings of Momotaz and Banik (2021) and Mantilla et al. (2025) further reinforce that seasonal price volatility and supply unpredictability require proactive digital solutions, justifying the inclusion of the Price Validity and Reconfirmation module.

On the subject of event booking and coordination, studies by Kusumawardhani et al. (2025), Deepthi et al. (2025), Vimalarani and Sathiyaseelan (2021), and JosephNg et al. (2023) consistently show that centralized, role-based digital booking platforms reduce scheduling conflicts, eliminate manual coordination errors, and improve service reliability in small-scale operations. Local studies by Vega and Generoso (2023) and Calucin et al. (2024) further demonstrate that this approach is applicable and effective within the Philippine SME context, directly informing the design of Raflora's Event Booking Coordination module.

On the subject of AI-assisted image analysis, research by Li et al. (2024), Onyejiaku et al. (2024), Sundaravadivel et al. (2025), and Wani et al. (2023) establishes that multimodal AI and computer vision systems can accurately recognize floral materials from visual input and translate those findings into structured operational data. The Philippine-context studies of Oyangoren and Camilo (2025), Balba et al. (2024), and Albano et al. (2025) further confirm that AI adoption in local SMEs is feasible but requires user-centered design, staff training, and clear validation workflows all of which are addressed by the system's human-in-the-loop administrator validation step.

This convergence of literature reveals a critical gap in existing platforms: while individual modules for event booking, perishable inventory management, and AI image analysis have each been developed and validated independently, no unified system has yet integrated all three functions within the specific operational context of a Philippine floral SME. This gap forms the primary technical justification for the proposed system.

Research Gap

While existing research has developed independent modules for event booking, inventory management, and AI image analysis, these technologies typically operate in isolation. Foreign studies on perishable inventory focus on shelf-life monitoring, while local Philippine studies emphasize general web-based coordination for SMEs. However, there is no existing integrated platform that links event booking, perishable-aware inventory, and AI-image analysis with specific seasonal substitution logic for the floral industry.

Furthermore, current systems lack a Price Validity and Reconfirmation module that connects AI-suggested materials with real-time pricing updates for long-term event bookings. This study fills this gap by developing a unified platform that explicitly links event data, seasonal rules, and AI-assisted analysis into a single ecosystem designed specifically for the operational constraints of small Philippine floral enterprises.

CHAPTER 3**RESEARCH METHODOLOGY****3.1 Research Method**

This study used the Rapid Application Development (RAD) Model as the software development methodology for the design and development of the Flower Arrangement Event Booking and Inventory Management System with Gemini AI-Assisted Image Analysis for Raflora Enterprises. RAD was chosen because it supports close collaboration with Raflora's owner and staff, allowing ongoing

refinement of each module to align with the operational workflows identified during requirements planning. Unlike sequential methodologies, RAD allows for evolving design details and facilitates rapid prototyping and user feedback in each development cycle. This approach is particularly suitable for a system that integrates an external AI component (the Gemini AI API) requiring iterative testing and adjustments during development. The system's specific focus covering a single-branch operation with three user roles and six core modules further supports the use of RAD, enabling researchers to progress through the requirements, design, and construction phases effectively without significant scope changes.

RAD Model

Rapid Application Development (RAD)

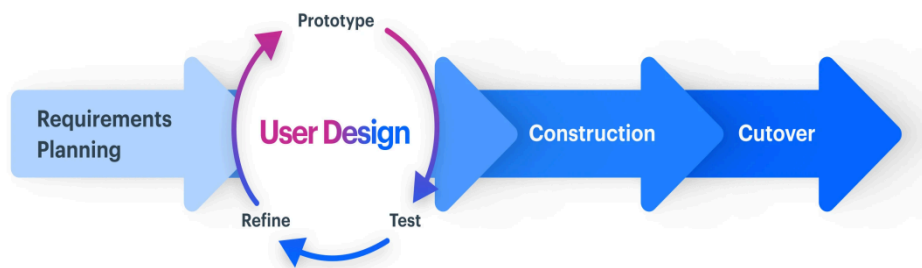


Figure 1: Rapid Application Development (RAD) Model

Phase 1: Requirements Planning

The Requirements Planning phase focused on identifying the operational problems, functional requirements, and system boundaries for the proposed system. The researchers conducted a semi-structured interview with the owner of Raflora Enterprises to gather first hand accounts of existing manual booking and inventory management workflows. A structured questionnaire was subsequently sent to the owner and designated staff members to systematically collect quantifiable data on recurring problems, including booking processing times, frequency of inventory shortages, pricing challenges, and desired system features. Direct observation of the company's manual operations, together with analysis of existing operational documents such as sample booking records and inventory logs, was used to define required system inputs, outputs, and data structures. The findings from this phase

were consolidated into a System Requirements Specification that served as the foundation for the subsequent User Design and Construction phases.

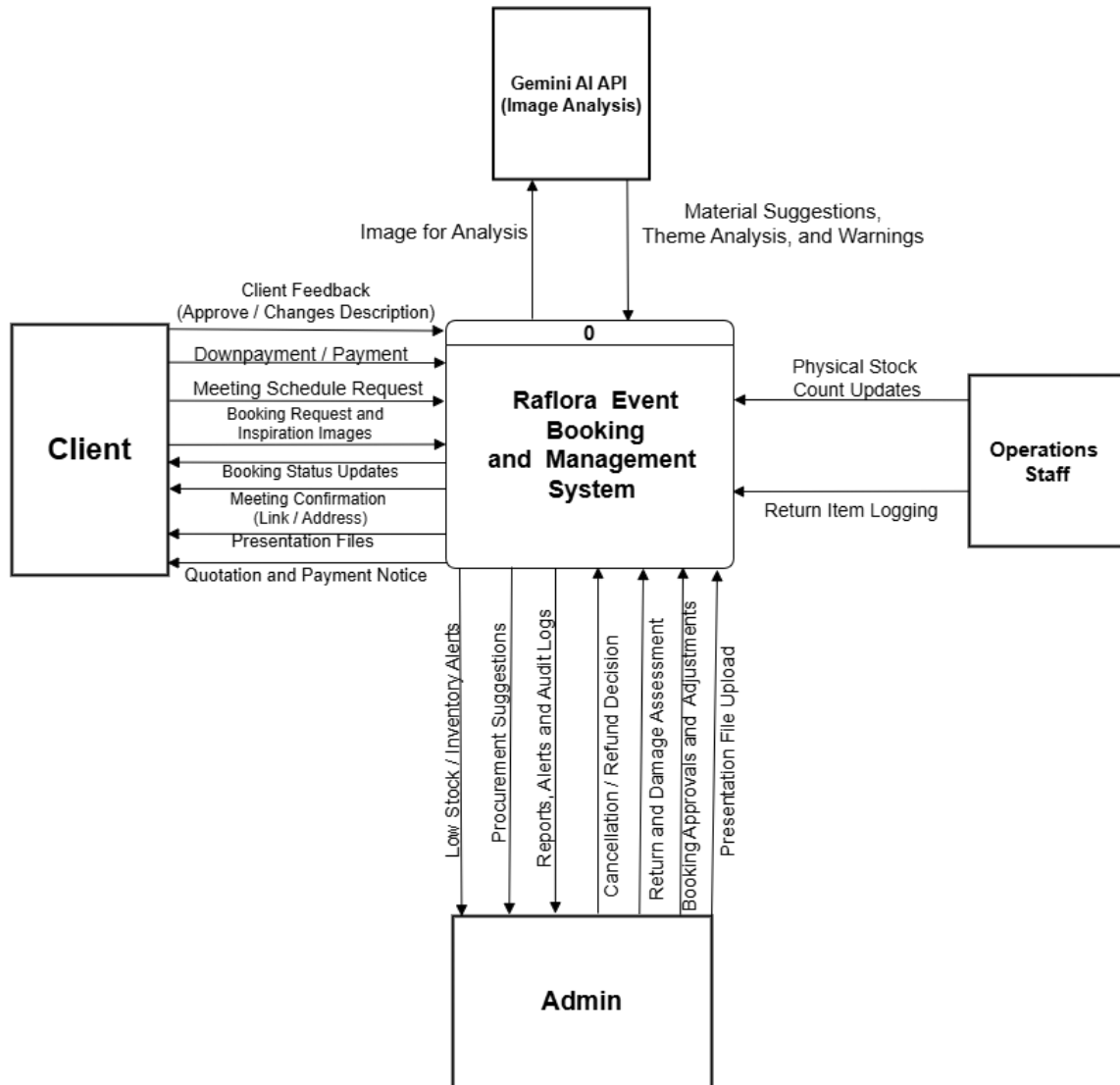


Figure 2: Context Diagram of the Proposed System

The Context Diagram shows the proposed system as a single high-level process interacting with four external entities: Client, Administrator, Staff, and Gemini AI API. The Client submits booking requests and inspiration images, receiving confirmations and price notifications. The Administrator provides approvals and inventory updates, receiving shortage alerts and reports; Staff submits inventory counts and return reports, receiving event checklists; and the Gemini AI API processes images to return material suggestions cross-referenced with inventory.

Phase 2: User Design

User Design, the project moves from high-level requirements into detailed technical modeling to create a blueprint for the system's architecture. This phase is centered on the development of the Entity Relationship Diagram (ERD), which defines the database schema and ensures all data from user accounts to AI-generated reports is logically structured and related. Alongside the ERD, Data Flow Diagrams (DFD) Levels 1 and 2 are constructed to map out the internal logic and specific processes, such as how the system handles image analysis and booking workflows. By finalizing these designs, the project establishes a clear roadmap that minimizes errors and streamlines the transition into the construction and coding stage.

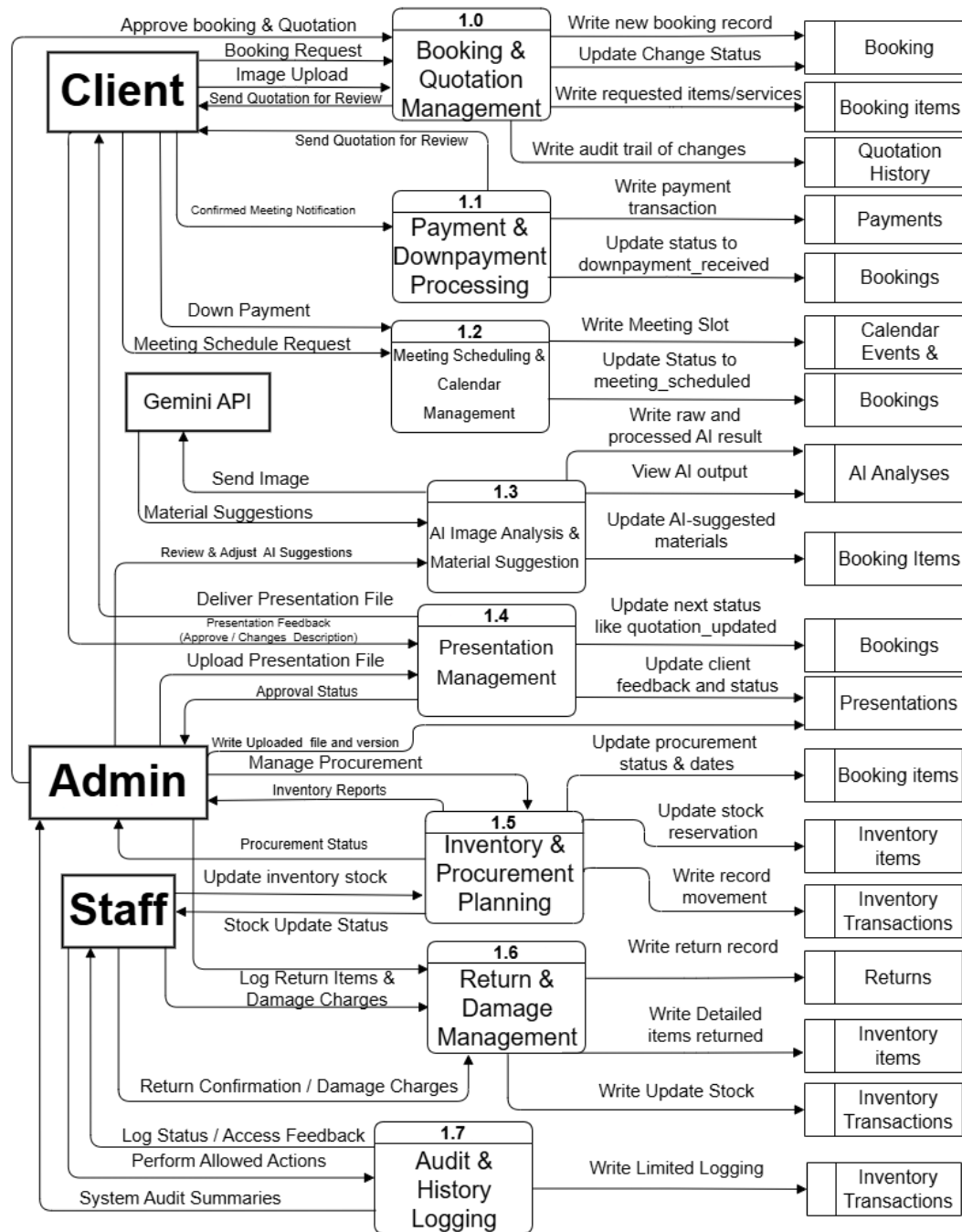


Figure 3: Data Flow Diagram level 1 of the Proposed System

Presents a Level 1 Data Flow Diagram (DFD) that elaborates on the detailed processes within the Flower Arrangement Event Booking and Inventory Management

System. The diagram breaks down the system into its primary functional modules: User Management, Booking Processing, Inventory Tracking, and AI Analysis. The Administrator initiates system configurations and inventory updates, while the Customer provides event requirements and reference images for processing.

The system records operational data into specific data stores, including user credentials, event booking records, and floral inventory levels. The AI Analysis process retrieves uploaded image data and communicates with the Gemini API to identify material requirements, which are then cross-referenced with the inventory data store to automate stock deduction. These data flows among the entities and processes illustrate a comprehensive booking-to-inventory cycle, emphasizing the transition from manual record-keeping to a synchronized digital infrastructure.

Data Flow Diagram Level 2

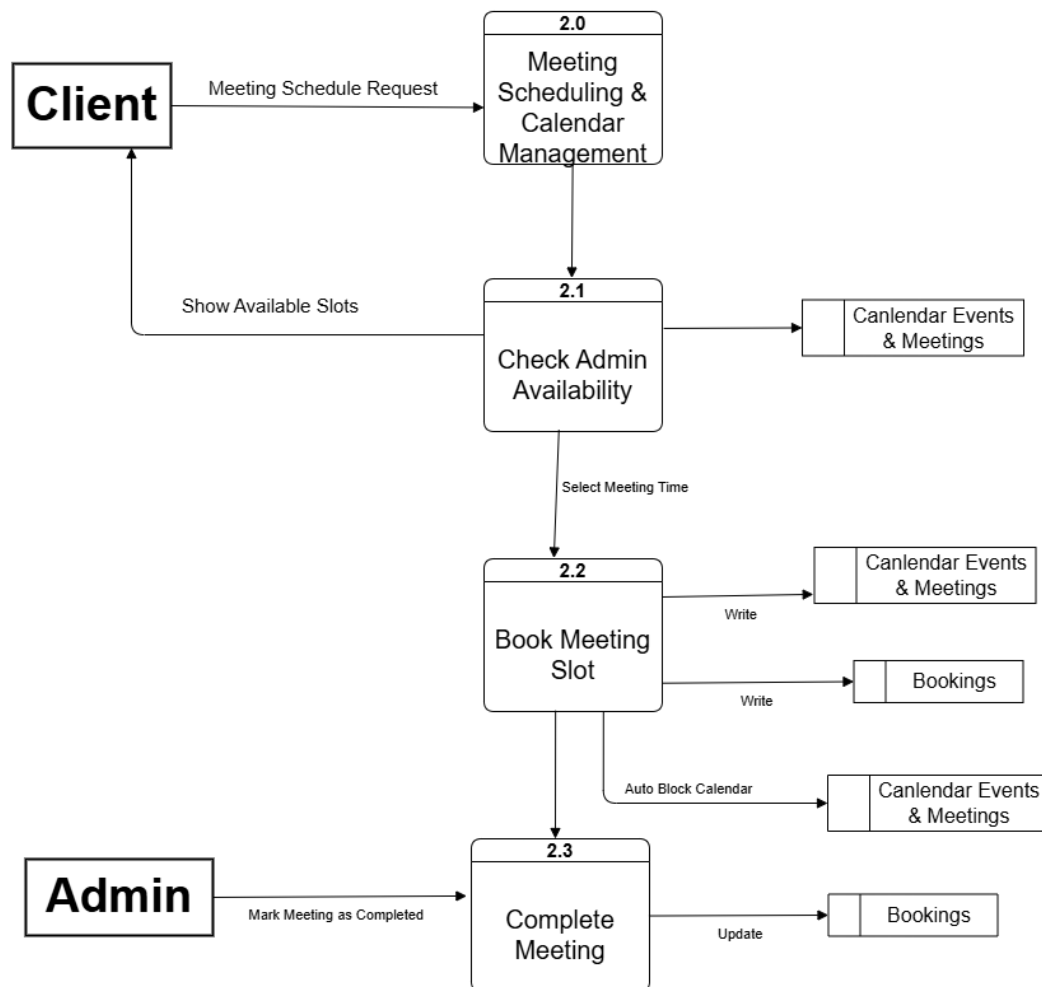


Figure 4: Data Flow Diagram Level 2 of the Proposed System

Decomposes the coordination process (Process 1.2) to show how the system manages availability. It outlines the interaction where the Client requests a schedule and views available slots retrieved from the Calendar Events and Meetings data store. Upon selecting a time, the system writes the booking and automatically blocks the

calendar, while the admin eventually marks the meeting as completed to update the final Bookings records.

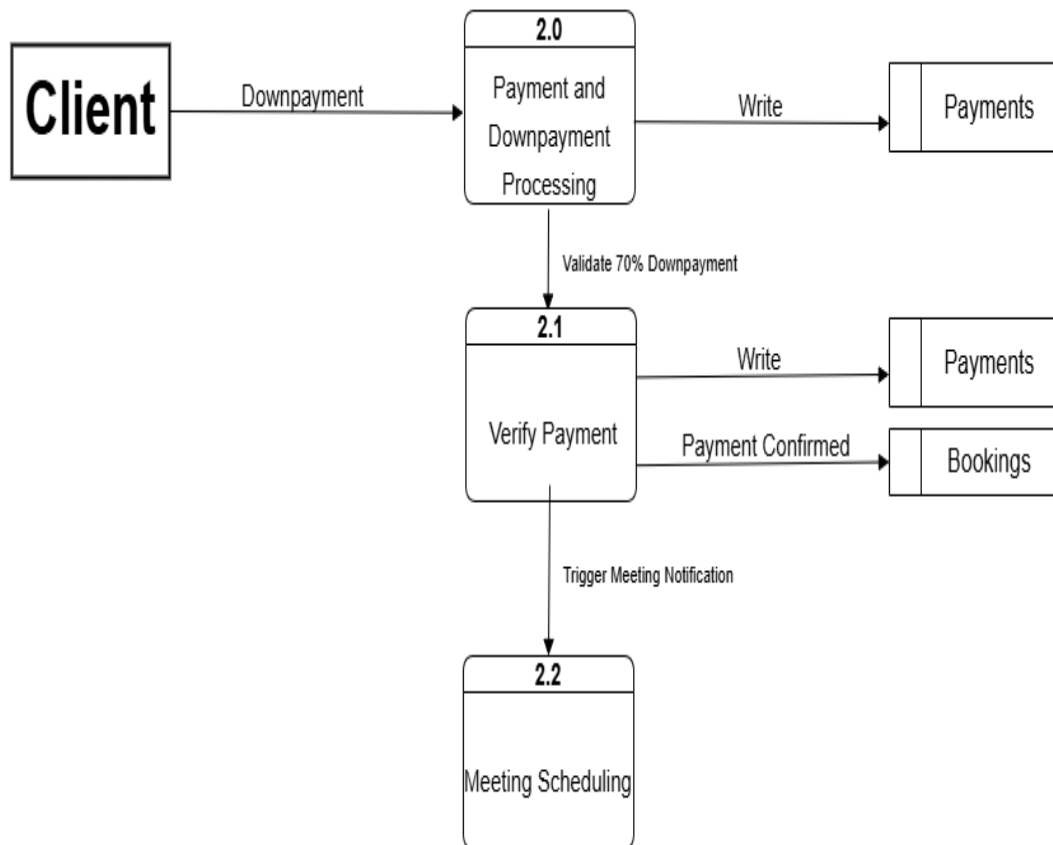


Figure 5: Data Flow Diagram Level 2 of the Proposed System

Illustrates the financial verification flow within Process 1.1 of the system. It captures the Client providing downpayment details, which are then written to the Payments data store. The system validates the required 70% downpayment to confirm the booking and triggers a notification to proceed with meeting scheduling once the payment status is verified.

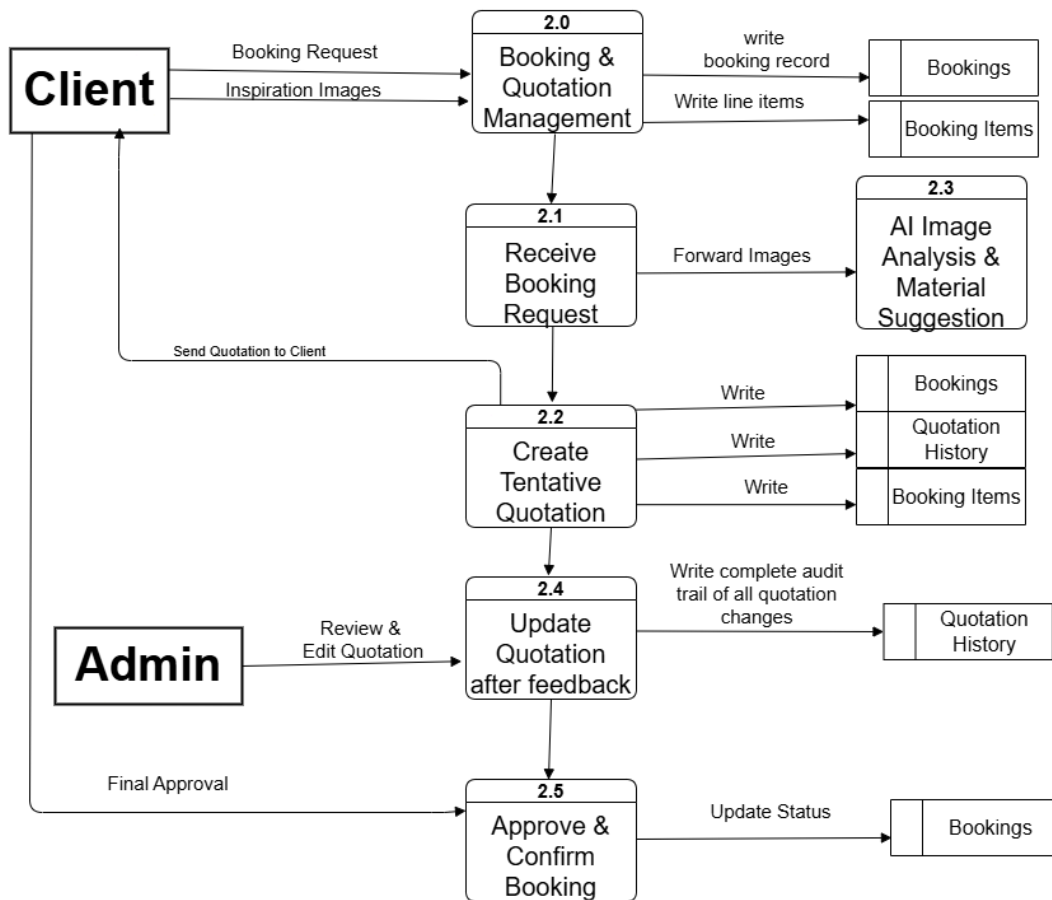


Figure 6: Data Flow Diagram Level 2 of the Proposed System

Presents the Level 2 DFD for Process 1.0, detailing the lifecycle of a booking request from initial inquiry to final confirmation. The process begins when the Client submits a booking request and inspiration images, which are processed to create a tentative quotation. The Admin reviews and edits these quotations based on feedback, maintaining a comprehensive audit trail in the Quotation History data store until final approval updates the Bookings status.

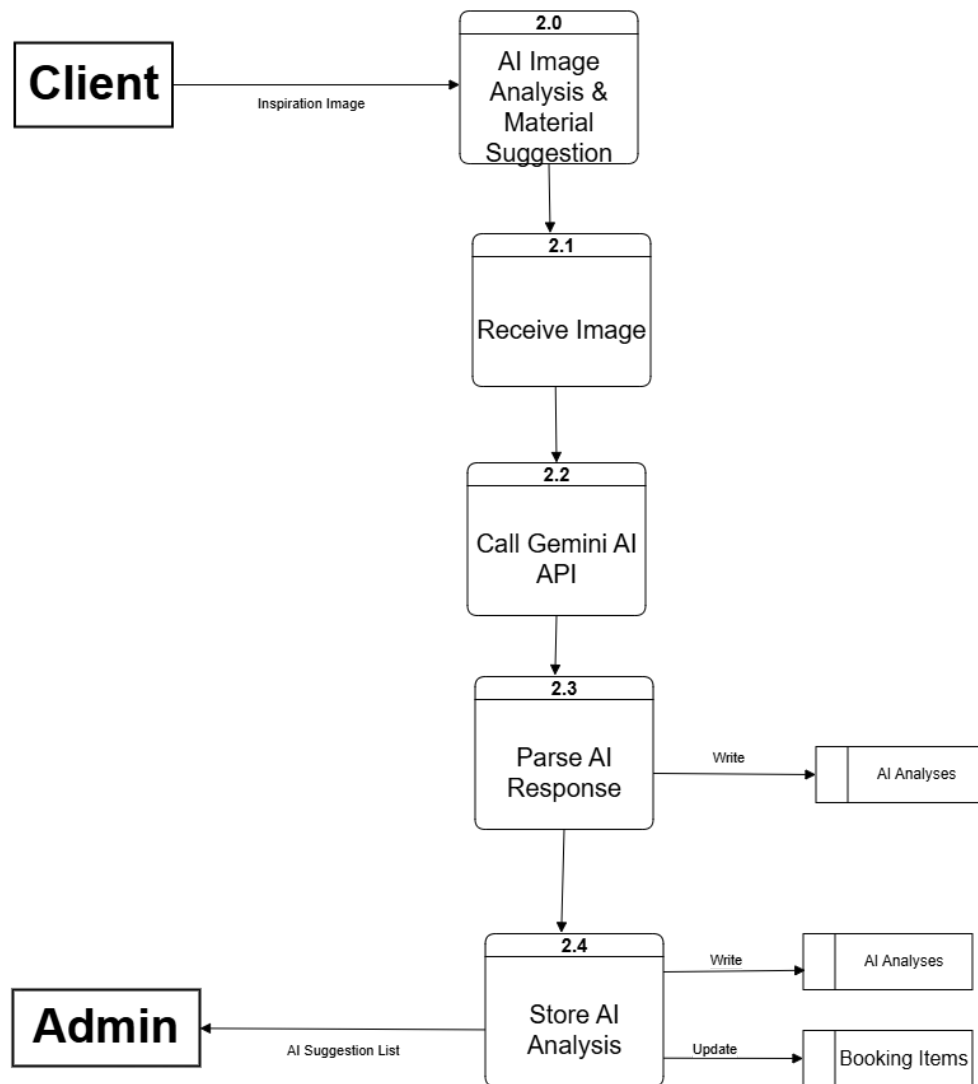


Figure 7: Data Flow Diagram Level 2 of the Proposed System

Shows the decomposed sub-processes of Process 1.3, which automates the identification of floral requirements. The flow moves from receiving the inspiration image to calling the Gemini AI API and parsing the response. The results are stored in the AI Analyses data store for Admin review, allowing for a validated update to the Booking Items that eliminates the need for manual material listing.

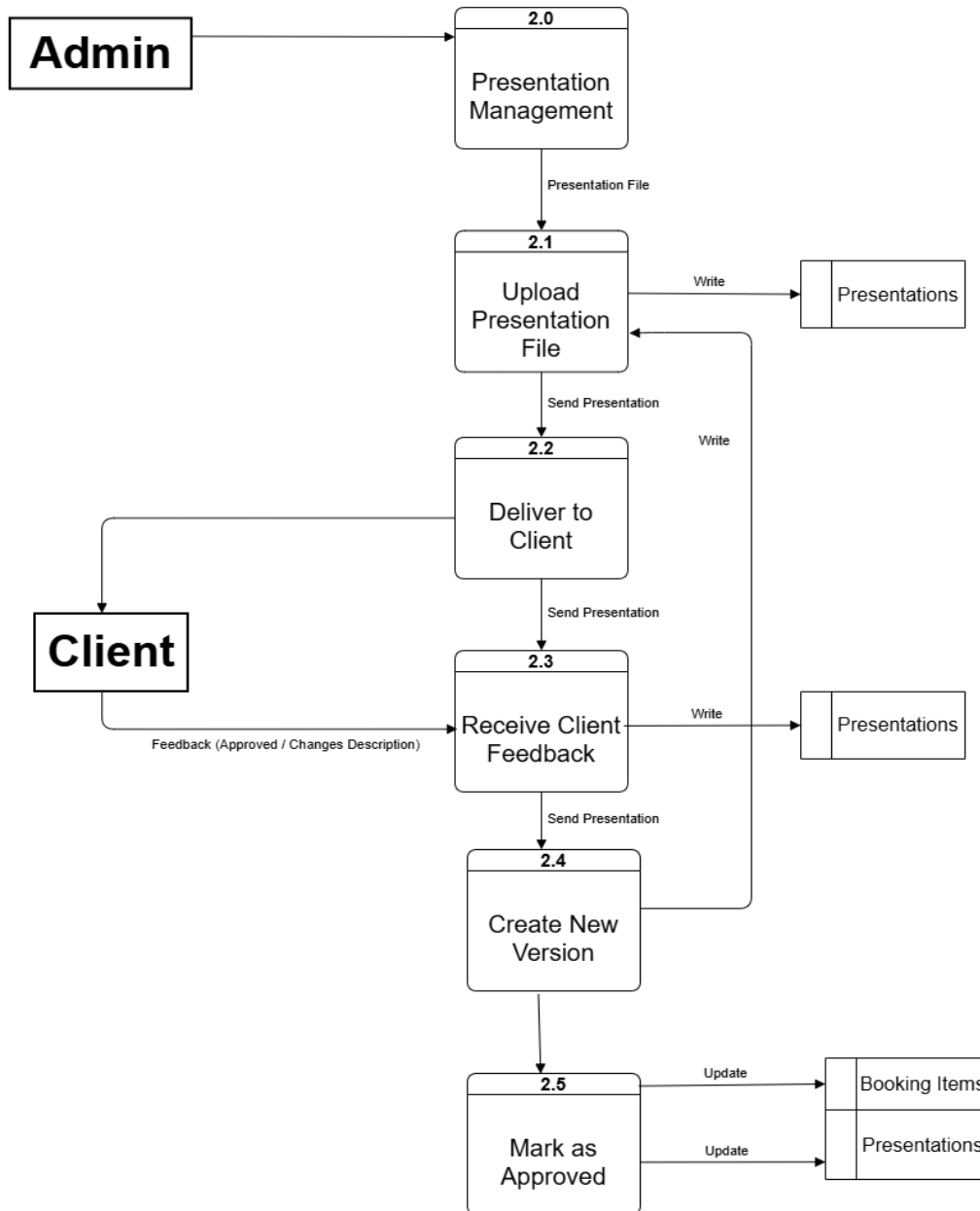


Figure 8: Data Flow Diagram Level 2 of Presentation Management

Describes the iterative design phase in Process 1.4 where the Admin manages visual proposals. The Admin uploads presentation files which are delivered to the Client for feedback. The system supports version control by creating new versions in

the Presentations data store based on client comments until the design is marked as approved, updating the final Booking Items.

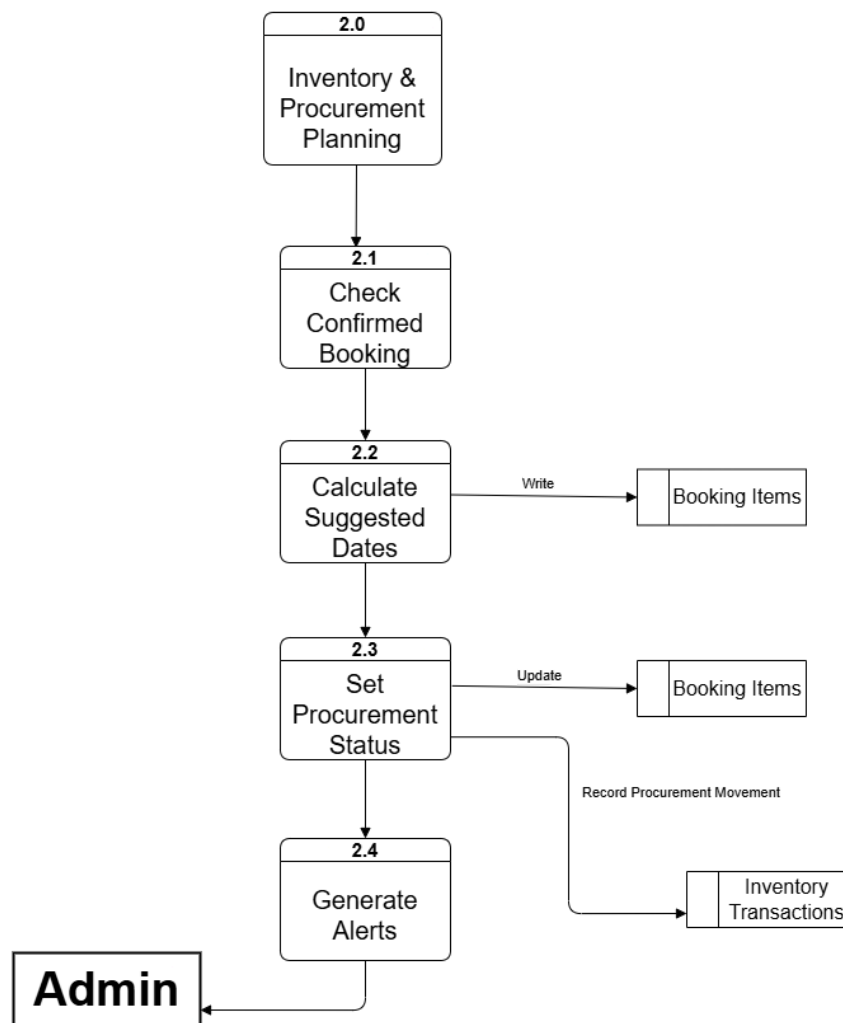


Figure 9: Data Flow Diagram Level 2 of the Proposed System

Elaborates on Process 1.5, focusing on the transition from confirmed bookings to logistical readiness. The diagram details how the system checks confirmed bookings to calculate suggested procurement dates for fresh stock. It records procurement

movements in Inventory Transactions and generates automated alerts for the Admin to ensure all materials are set for the event date.

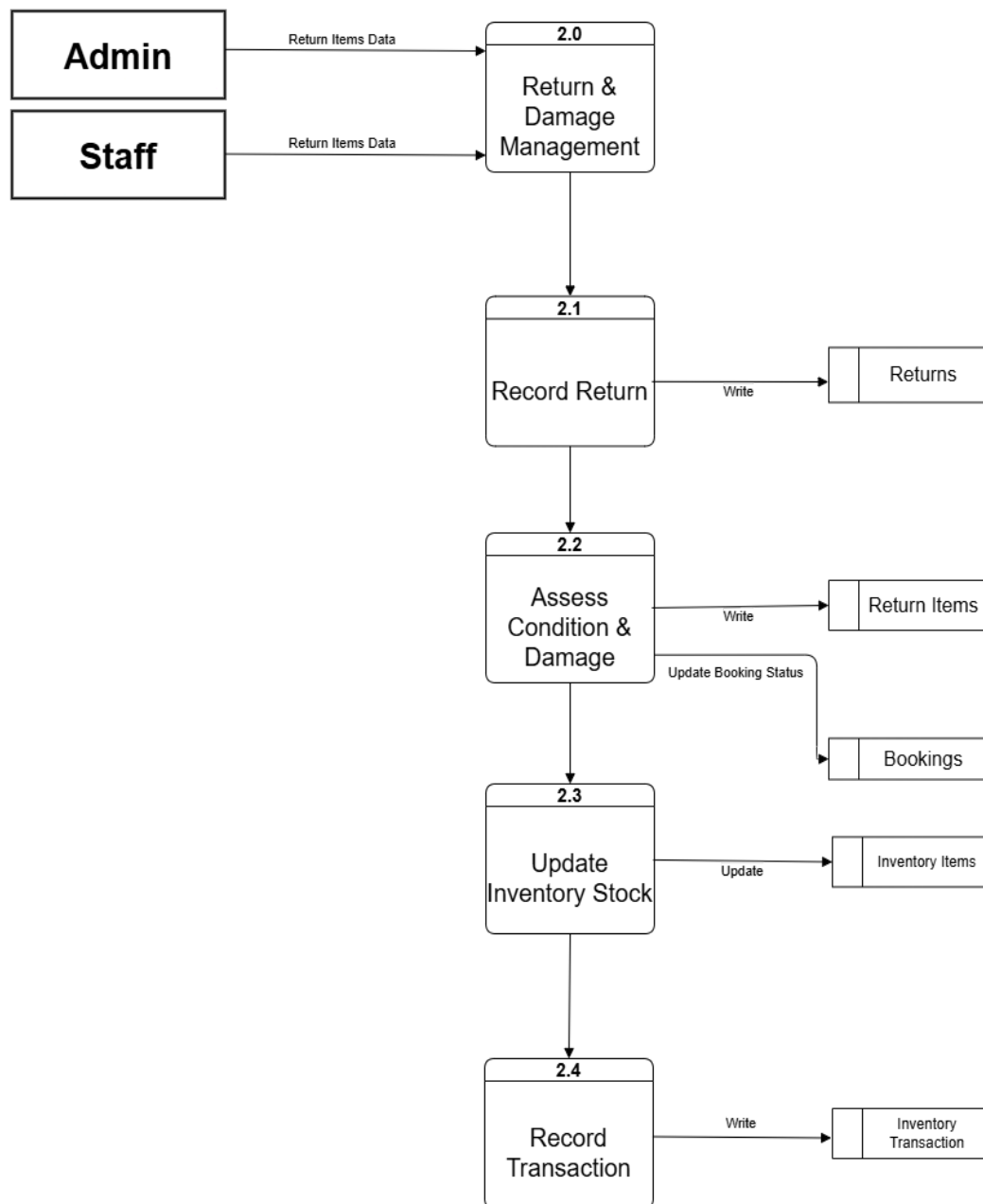


Figure 10: Data Flow Diagram Level 2 of the Proposed System

Describes the post-event logistical flow (Process 1.6) for handling equipment returns and assessing damages. The process accepts return data from Admin or Staff, recording the entry before assessing item conditions. Data is written to Returns and Return Items, ultimately updating the Inventory Items stock levels and recording the final movement in the Inventory Transaction store.

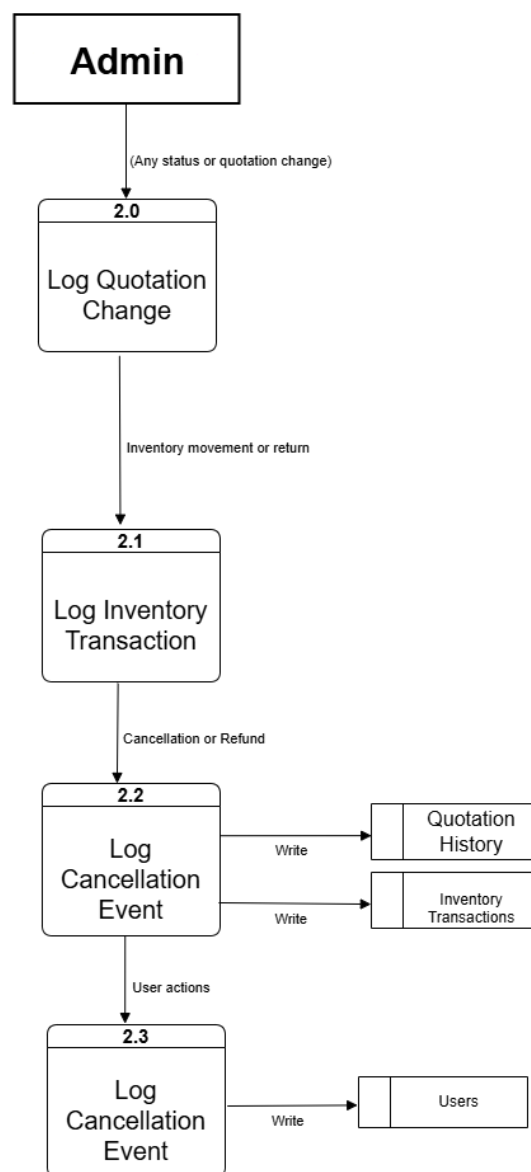


Figure 11: Data Flow Diagram Level 2 of the Proposed System

Presents the Level 2 DFD for Process 1.7, which ensures system accountability and data integrity. It shows how the Admin triggers logging for various events, including quotation changes, inventory transactions, and cancellations. These actions are written to the Quotation History, Inventory Transactions, and Users data stores, providing a transparent record of all critical system activities and user actions.

Phase 3: Construction

The Construction phase focuses on building and testing the system modules through a step-by-step process. The application will be developed using the Laravel framework for the backend, HTML, CSS, and Bootstrap for the user interface, and MySQL for managing the database. A major part of this phase is the integration of the Gemini AI API, which allows the system to analyze inspiration photos uploaded by clients and suggest the specific floral materials needed for an event. To help with shop operations, the system includes Automated Scheduling Logic that monitors event dates and stock levels. This logic is programmed to send out alerts that start four weeks before a confirmed event and move to daily reminders during the final week to ensure all materials are secured. The development follows a logical sequence starting with user logins and event booking, moving to inventory and AI analysis, and ending with price checks and reports. Each module undergoes careful testing to ensure that the entire system works as a unified and reliable tool for Raflora Enterprises.

Phase 4: Cutover

The Cutover phase will involve the deployment of the completed system into a testing environment accessible to Raflora Enterprises' administrators and staff, followed by User Acceptance Testing (UAT) and system orientation. System orientation sessions will be conducted with designated respondents to familiarize them with the system's navigation, role-specific features, and module workflows prior to formal evaluation. UAT will be conducted using a standardized software quality

evaluation instrument adapted from the ISO/IEC 25010 Software Quality Model to assess the system's practical functionality and overall quality in the context of the company's actual event coordination and inventory management operations. Post-deployment activities within this phase will include monitoring system behavior under actual usage conditions, resolving defects identified during evaluation, and implementing minor enhancements based on respondent feedback. Performance metrics specifically booking processing time, incidence of scheduling conflicts, and inventory alert response accuracy will be recorded during this phase to support the comparative evaluation.

3.2 Conceptual Framework

INPUT	PROCESS	OUTPUT
Data Requirements: Client-Provided Data: Event details, booking dates, and contact information. Visual Input: Client-uploaded inspiration images (JPG/PNG). Inventory Data: Current floral stock levels, unit costs, and material categories. Operational Parameters: Price validity periods, seasonal substitution rules, and	System Logic and Transformation: AI-Assisted Image Parsing: Analyzing visual data via the Gemini AI API to identify floral materials. Booking Coordination: Validating event dates and managing the reservation lifecycle. Inventory Synchronization: Real-time stock deduction and shortage detection logic.	System Deliverables: A Unified Web-Based Platform: A Flower Arrangement Event Booking and Inventory Management System with Gemini AI-Assisted Image Analysis AI-Generated Material Suggestions: Structured lists of recommended flowers and quantities. Optimized Event Schedules: A

non-perishable asset lists. User Credentials: Administrator and staff authentication data.	Price Reconfirmation: Triggering alerts for tentative quotations based on date thresholds. Asset Recovery Tracking: Logging the return and status of non-perishable materials. Data Aggregation: Generating audit logs and operational reports through MySQL queries.	conflict-free digital calendar of bookings. Real-Time Inventory Alerts: Automated notifications for low stock and procurement needs. Operational Reports: Timestamped audit logs, material usage summaries, and business performance analytics.
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Table 1: Input Process Output (IPO) Model

The conceptual framework follows the Input-Process-Output (IPO) model to illustrate the system's operational flow. The Input phase captures unstructured data from event images and manual booking details, which are then transitioned into the Process phase for AI-assisted material extraction and database synchronization. To ensure reliability, the processing stage includes a human-in-the-loop validation step where administrators confirm AI-generated suggestions before they are committed to the database. This technical workflow ensures that high-accuracy data is fed into the Output phase, which generates structured results such as automated material requirements and a conflict-free digital calendar. By standardizing these procedures, the framework effectively bridges the visibility gap between Raflora Enterprises' bookings and inventory stock.

3.3 Data Gathering Tools and Procedures

The researchers employed four data gathering methods to support the Requirements Planning phase of the RAD methodology. These activities identified Raflora Enterprises' operational problems, workflows, and system requirements detailed in Section 1.1.

Data Gathering Tools

Interview. A semi-structured interview was conducted with Raflora Enterprises' owner to document manual booking and inventory processes. The session confirmed operational problems from Section 1.1 and identified requirements for booking coordination, inventory alerts, role-based access, and audit logging.

Questionnaire. A structured questionnaire distributed to the owner and five designated staff members validated interview findings. It collected preferences for inventory alert frequency, AI material suggestions, and role-restricted financial data access.

Observation. Direct observation at Raflora facilities verified manual processes including notebook updates, wall calendar maintenance, physical shelf counts, and post-event material tracking.

Document Analysis. Existing booking records, inventory logs, and quotation documents were analyzed to define system inputs, outputs, and database structures for all six modules.

Data Gathering Procedures

The researchers employed a multi-phased data gathering approach to ensure a holistic understanding of the business requirements. The process began by securing formal written permission from the owner of Raflora Enterprises to conduct research activities.

Initially, the researchers conducted semi-structured interviews and distributed questionnaires to the owner and five staff members. These tools were designed to document existing workflows, validate the problems identified in the project context, and collect specific user preferences for the system's features. Following the interviews, the research team performed on-site observations at the Raflora facilities. These sessions allowed the researchers to verify manual processes in real-time and document the specific time required to cross-reference flower availability, confirming a manual processing delay of approximately one to two hours.

Finally, operational documents were collected and analyzed to identify precise data requirements. All gathered qualitative and quantitative findings were consolidated, which served as the technical foundation for the User Design phase of the RAD methodology."

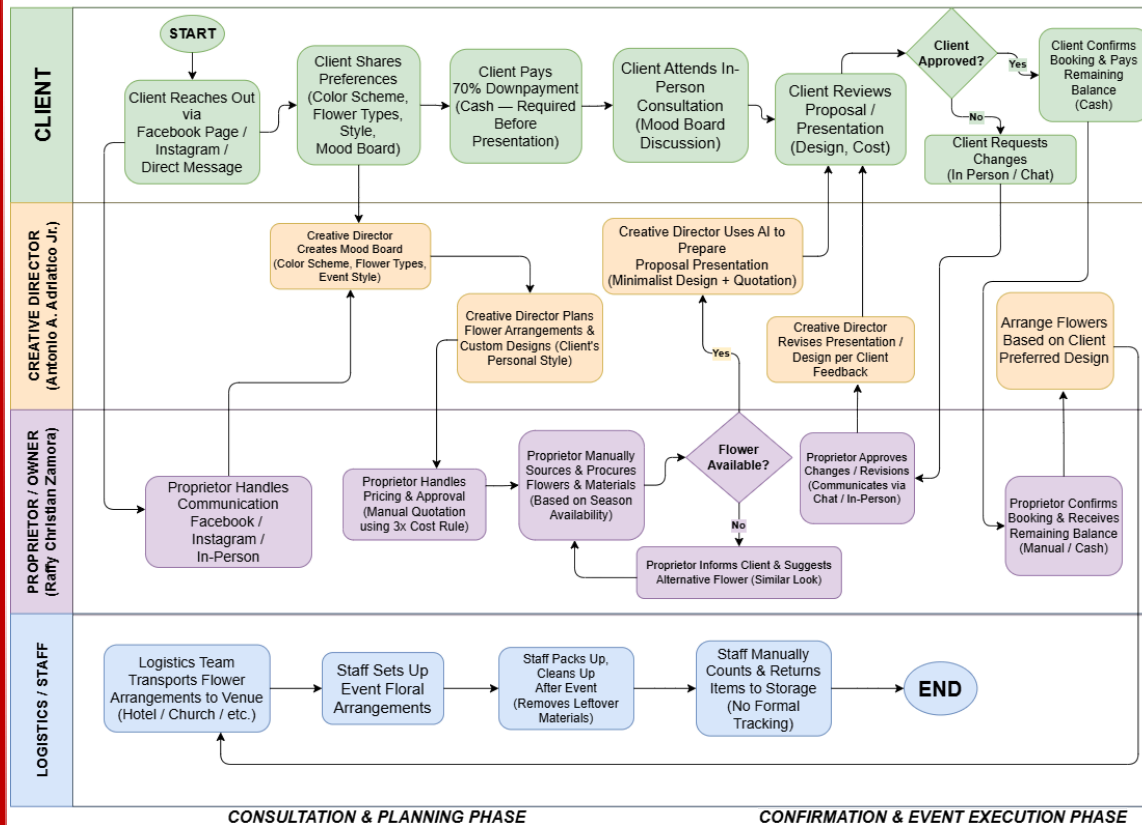


Figure 12: Flowchart of Existing System

Presents the Activity Flowchart of the Existing System, illustrating the manual and paper-based operational sequence currently utilized by Raflora Enterprises. The diagram identifies a linear, decentralized process where event inquiries and inventory checks are conducted through physical media and fragmented communication channels. The workflow begins with a manual client inquiry and moves through a series of human-dependent steps, including physical stock verification and handwritten booking entries in a traditional logbook.

The process logic reveals a lack of automated synchronization, as data is recorded into non-relational formats that lack centralized validation. Because there is no

integrated digital hub, the flow illustrates significant bottlenecks where staff must manually cross-reference event requirements against physical inventory lists. This visualization emphasizes the systemic risks of data redundancy and record loss inherent in the manual-decentralized model, providing the technical justification for the proposed digital transformation.

3.4 Project Design

The Project Design is based on the Rapid Application Development (RAD) methodology, which emphasizes an iterative process between the researchers and the system's end-users. The diagram illustrates the core development lifecycle, moving from Requirement Planning to the User Design and Construction phases. This design justifies the accelerated development timeline by showing how user feedback is directly integrated into the prototyping process to refine the system's features and interface.

The logic of this design ensures that the technical requirements such as the AI-assisted image analysis and inventory modules are validated early in the development cycle. By maintaining a continuous feedback loop, the project design minimizes the risk of logical gaps between the proposed software and the actual operational needs of Raflora Enterprises. This iterative framework serves as the roadmap for the system's architecture, ensuring a final product that is both functionally robust and aligned with the users' technical capabilities.

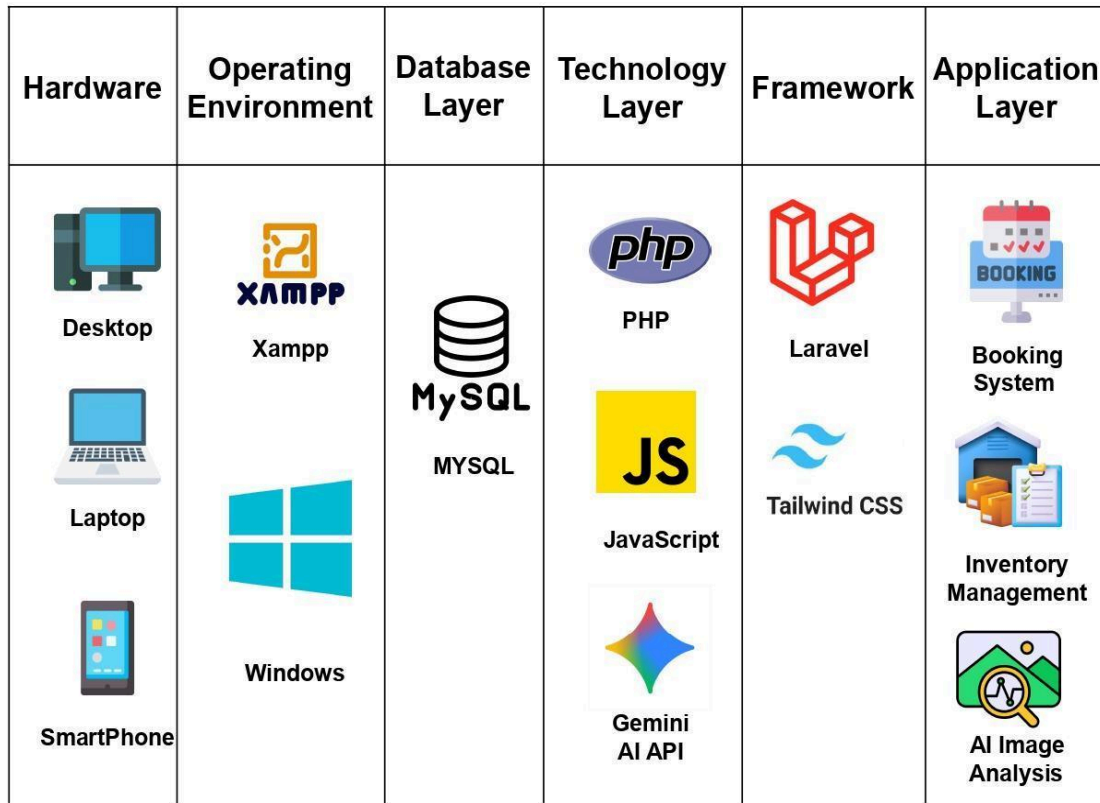


Figure 13: System Architecture of the Proposed System

Presents the System Architecture, illustrating the structural relationship between the hardware components, the software interface, and the backend cloud services. The diagram defines a three-tier architecture consisting of the Client Layer (Web Interface), the Application Layer (Logic and AI Parsing), and the Data Layer (MySQL Database). This architectural logic demonstrates how user inputs from the local interface are transmitted to the Gemini AI API for material identification before being stored in the centralized database.

The architecture emphasizes the secure and efficient flow of data between the internal system and external cloud environments. By utilizing an API-driven design, the system offloads complex computational tasks to the Gemini suite while maintaining a

lightweight local footprint for the Administrator. This structure justifies the system's ability to provide real-time inventory updates and booking management, establishing a high-performance foundation that eliminates the processing latencies found in the previous manual-decentralized setup.

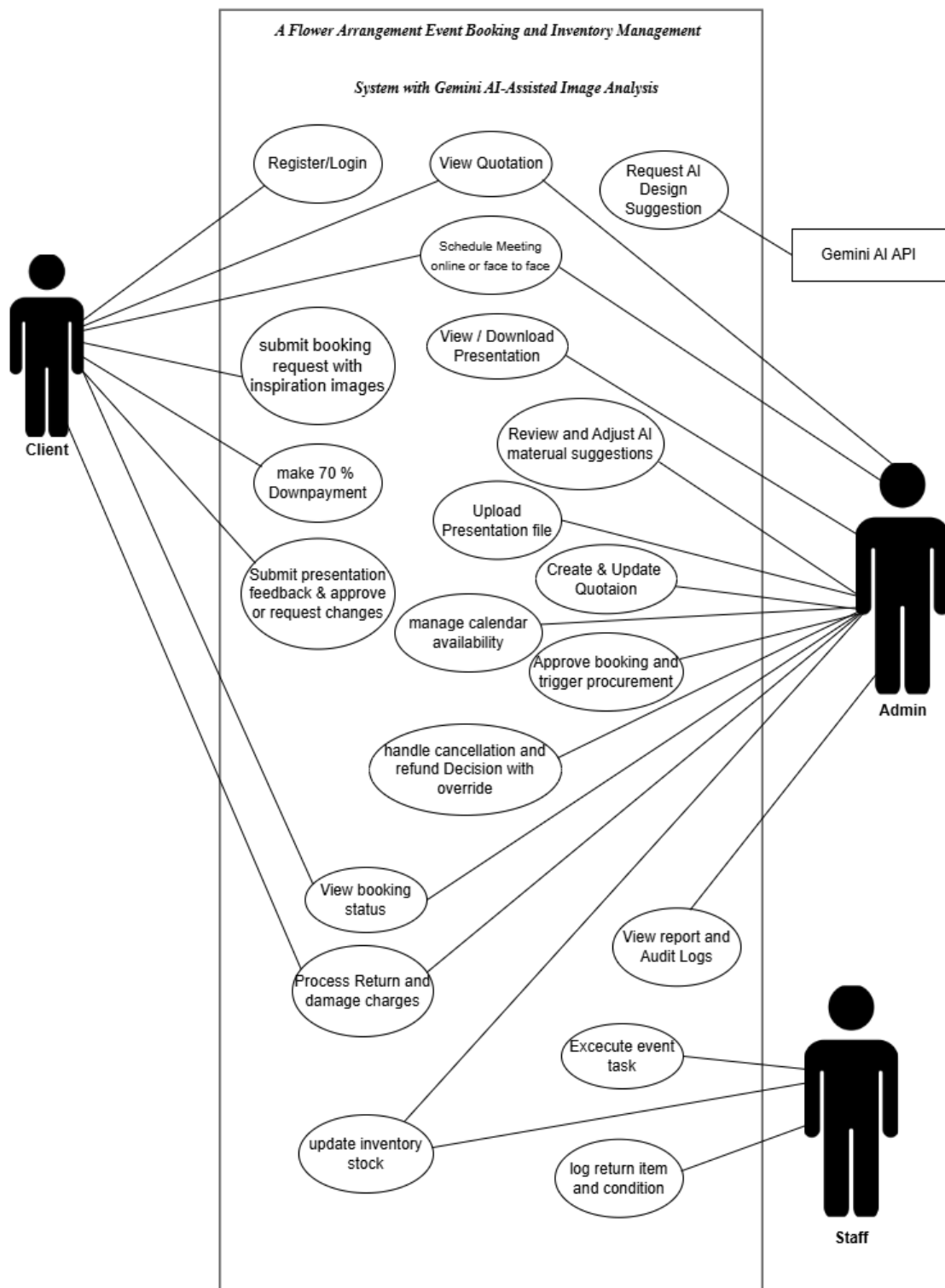


Figure 14: Use Case Diagram of the Proposed System

Presents the Use Case Diagram, which defines the functional boundaries of the system and the specific interactions between the primary actors and the platform's core modules. The diagram identifies the Administrator as the primary actor and the Gemini API as the secondary actor. The architectural logic centers on the "Event Booking" and "Inventory Management" use cases, which serve as the primary triggers for the system's automated features.

The process logic illustrated in the use case highlights the Gemini AI-Assisted Image Analysis as a critical relationship for the booking process. When the Administrator initiates a new event booking and uploads a reference photo, the system invokes the Gemini API to parse visual data into a structured material list. This interaction justifies the system's ability to automate material identification, effectively bridging the gap between client requirements and physical stock levels. By defining these boundaries, the Use Case Diagram serves as a behavioral blueprint ensuring that all identified operational problems such as manual counting errors and decentralized data are addressed through specific system-user interactions.

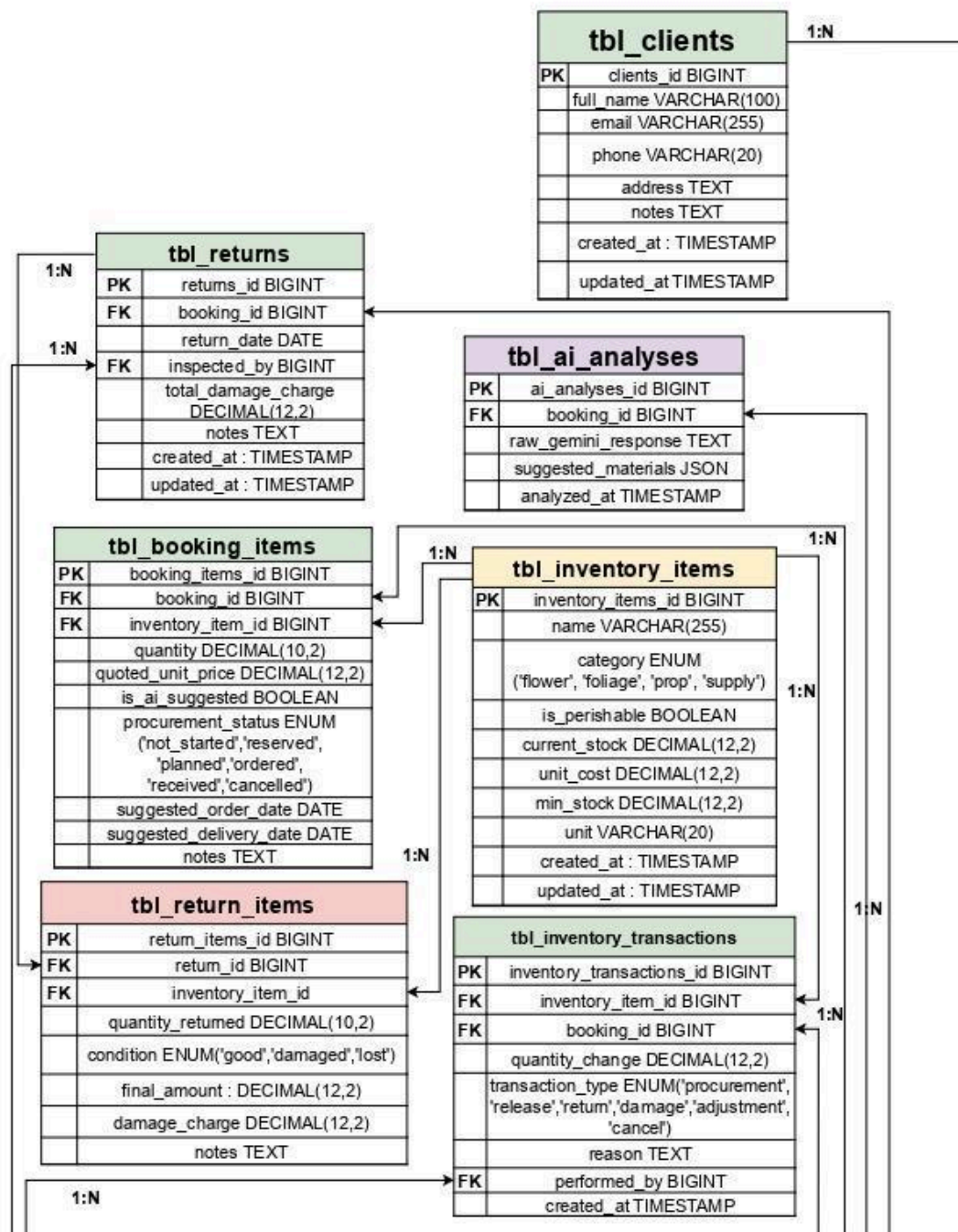


Figure 15: Entity Relationship Diagram of the Proposed system (Part 1)

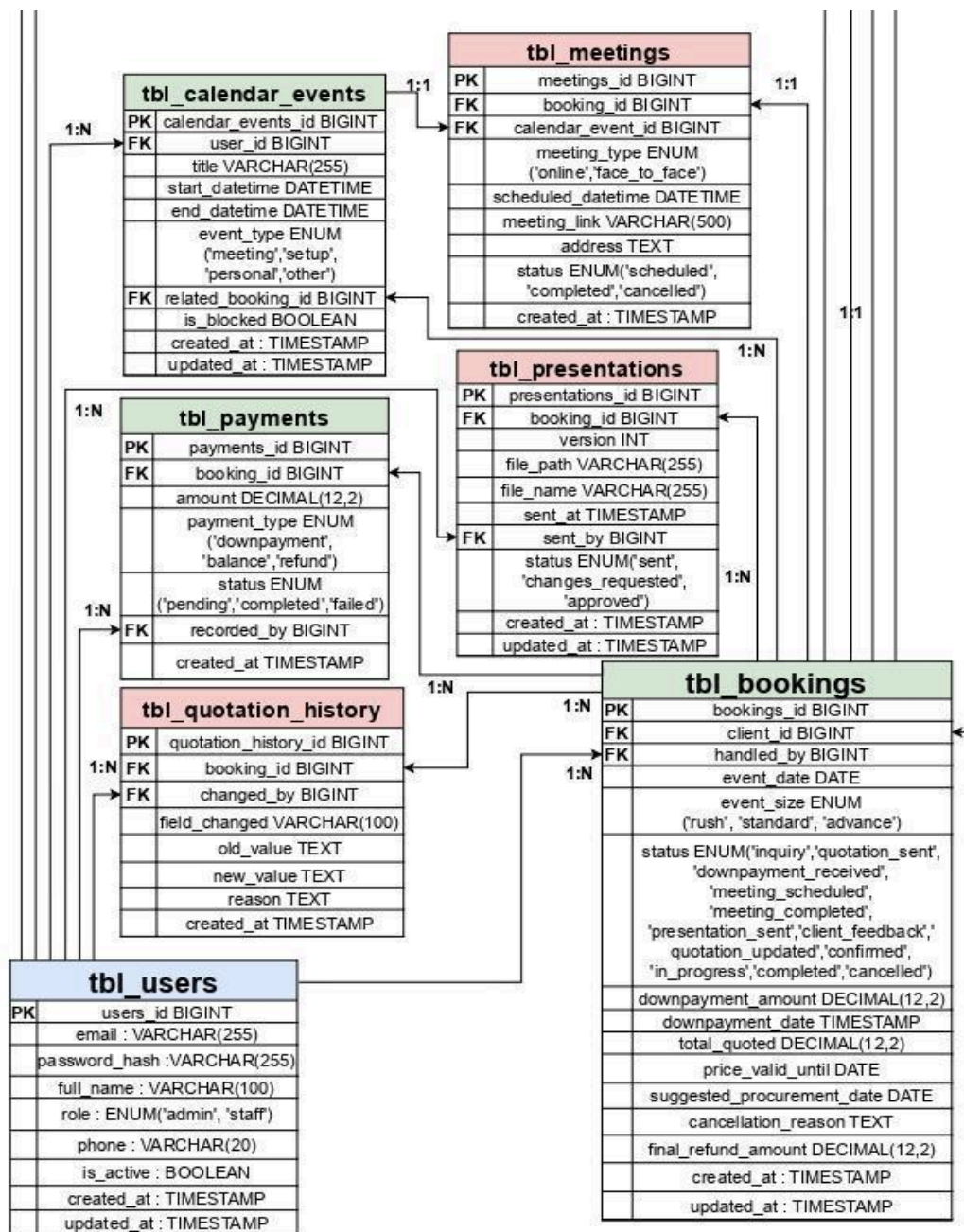


Figure 16: Entity Relationship Diagram of the Proposed system (Part 2)

Presents the Entity Relationship Diagram (ERD), which illustrates the relational database schema and the logical structure of the system's data storage. The diagram identifies core entities including Users, Bookings, and Inventory linked through

primary and foreign key constraints to ensure referential integrity. To manage the complex requirements of floral arrangements, an associative entity is utilized to resolve the many-to-many relationship between unique bookings and specific inventory items.

The database records transaction-level data, such as material quantities, unit costs, and event schedules, into normalized tables to prevent data redundancy. When a booking is finalized, the system executes an automated update across these related entities, ensuring that stock levels are deducted in real-time. This relational mapping emphasizes the transition to a centralized source of truth, providing the technical foundation necessary to eliminate the record loss and material discrepancies common in paper-based documentation.

3.5 Project Evaluation

The evaluation assessed usability and operational effectiveness during the RAD Cutover phase. Two approaches were used with Raflora staff: a survey questionnaire adapted from the ISO/IEC 25010 Software Quality Model evaluating the system across six quality characteristics: Functional Suitability, Usability, Reliability, Performance Efficiency, Security, and Compatibility; and a before-after performance comparison against the Section 1.1 baseline metrics of booking processing time, scheduling conflicts, and inventory shortage incidence.

Respondents of the Study

The respondents of the study consist of five (5) individuals, comprising the owner, the lead floral designer, and three administrative staff members of Raflora Enterprises. While this represents a small numerical sample, the researchers utilized a total population sampling technique. Since these five individuals constitute 100% of the administrative and decision-making workforce responsible for the booking and inventory management processes, they are the only qualified subjects capable of evaluating the system's impact on these specific internal workflows. While the business employs additional general staff for event setup, the selected respondents represent the total population of the system's intended end-users. Consequently, their feedback will provide a comprehensive and highly valid assessment of the system's usability and effectiveness within its intended organizational context.

Evaluation Design

The evaluation of the system is divided into three parts to test its usability, business impact, and AI accuracy. First, the researchers will conduct User Acceptance Testing (UAT) with five respondents representing the Administrator, Staff, and Client roles who will perform tasks like booking events and checking inventory. Their feedback will be measured using a survey questionnaire adapted from the ISO/IEC 25010 Software Quality Model, evaluating the system across six quality characteristics: Functional Suitability, Usability, Reliability, Performance Efficiency, Security, and Compatibility. Second, a performance analysis will compare the new system to the old manual way of working, specifically measuring how much time is saved and if scheduling errors are reduced. Finally, the AI Model Performance Validation will be conducted to check the accuracy of the Gemini AI-Assisted Image Analysis. The Ground Truth was established through a validation session conducted with the shop owner, wherein a set of client-provided inspiration images representing varying arrangement styles, color schemes, and event types will be pre-verified to identify the correct floral materials. The AI's suggestions for each image will then be compared against these verified materials to compute Precision, Recall, and User Agreement Rate.

Evaluation Instrument

Quality Characteristic	Definition
Functional Suitability	System performs tasks correctly and completely
Usability	Easy to learn, navigate, and operate
Reliability	Operates consistently without errors
Performance Efficiency	Responds within acceptable time limits
Security	Protects data and access from unauthorized use
Compatibility	Functions across web browsers and devices

Table 2: ISO/IEC 25010 Software Quality Model.

Ensuring a standard and objective assessment, the primary tool for data collection is a survey questionnaire adapted from the ISO/IEC 25010 Software Quality Model. This instrument evaluates the system based on six specific quality characteristics: Functional Suitability, Usability, Reliability, Performance Efficiency, Security, and

Compatibility. Each criterion is assessed using a five-point Likert Scale, with ratings ranging from 5 (Highly Accepted) to 1 (Not Acceptable). By utilizing this standardized instrument, the researchers can calculate the weighted mean of the responses, providing a statistically sound basis for determining the system's technical quality and its readiness for full-scale implementation.

Likert Scale

Scale	Range	Interpretation
5	4.21 – 5.00	Highly Acceptable
4	3.41 – 4.20	Acceptable
3	2.61 – 3.40	Neutral
2	1.81 – 2.60	Slightly Acceptable
1	1.00 – 1.80	Not Acceptable

Table 3: *Likert Scale and Interpretation*

Presents the Five-Point Likert Scale used to quantify the respondents' evaluation of the system's functional and non-functional requirements. The scale provides a standardized metric for measuring subjective qualitative feedback such as usability, reliability, and AI accuracy into objective numerical data. This conversion is essential for the statistical analysis phase, allowing the researchers to determine the overall

level of acceptance of the Flower Arrangement Event Booking and Inventory Management System.

The interpretive ranges applied to the computed weighted mean scores justify the system's readiness for deployment. By categorizing scores into specific levels of acceptability, the researchers can pinpoint which modules such as the Gemini AI-assisted parsing or the MySQL inventory synchronization meet the high standards of the end-users. This logical mapping ensures that the final evaluation is not based on general impressions but on a rigorous, data-driven assessment of the software's operational performance.

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